

Math 161

Maps, images, preimages

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Let X and Y be sets.

Pseudo-Definition. A *map* (or *mapping* or *function*) $f : X \rightarrow Y$ is a rule which assigns to any element $x \in X$ an element $f(x) \in Y$.

We will give a more precise definition in class, but this suffices for most purposes.

Given a map $f : X \rightarrow Y$, we call X the *domain* or *source* of f , and Y the *codomain* or *target*.

A note on terminology: traditionally the word “function” is used specifically for maps whose target consists of numbers. Most maps we encounter in calculus are of this kind, but for now we’ll use the term “map” to evoke a more general context.

Often maps are given by a single formula, such as the map $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) := x^2$. Sometimes they may be defined by several formulas (“piecewise” maps), or even by specifying where each element goes individually! For example, we could define a map as follows:

$$\begin{aligned}\{a, b, c, d\} &\longrightarrow \{1, 2, 3\} \\ a &\mapsto 2 \\ b &\mapsto 3 \\ c &\mapsto 2 \\ d &\mapsto 1.\end{aligned}$$

Note that the symbol $\rightarrow$ is used when declaring a map, while $\mapsto$ is used to specify where elements go. So a map $f : X \rightarrow Y$ could also be denoted by $x \mapsto f(x)$, although the latter notation leaves the domain and codomain unspecified.

Let $f : X \rightarrow Y$ be a map and $A \subset X$, $B \subset Y$ subsets. The *image* of A under f is

$$f(A) := \{f(x) \mid x \in A\} \subset Y,$$

while the *preimage* of B under f is

$$f^{-1}(B) := \{x \in X \mid f(x) \in B\} \subset X.$$

In the case $A = X$, we sometimes call $f(X)$ the *range* of f .